

ITS Camera Pixel Density Quick Reference

Recommended pixel density targets for Intelligent Transportation Systems (ITS) camera deployments.

Application	Pixels/ft	Purpose
Vehicle Presence	8–10	Basic vehicle detection & occupancy
Vehicle Count	10–15	Traffic volumes & turning counts
Classification	15–20	Cars, trucks, buses & motorcycles
Queue Detection	18–20	Congestion & adaptive timing
Wrong-Way	20+	Wrong-way alerts
Signal Actuation	20–25	Loop/radar replacement
Speed Estimation	20–25	Traffic speed studies
ALPR	45–60+	License plate recognition

Bosch Pixel Counter Example

The screenshot shows the Bosch FLEXIDOME 8100i - 2MP X series web interface. The main content area is titled 'Pixel Counter' and displays a live video feed of a street intersection. A green bounding box is drawn around a vehicle in the video. Below the video, the pixel dimensions for four streams are listed: Stream 1 (173 x 117 px), Stream 2 (138 x 93 px), Stream 3 (115 x 78 px), and Stream 4 (65 x 43 px). The interface also includes a sidebar menu with options like General, Web Interface, Connectivity, Camera, Recording, Alarm, Network, and Service. A live preview window on the right shows the video feed with a 'Stream 1' dropdown and a 'Live preview' button.

Example Calculation

If the Pixel Counter reports **173 pixels** across the full width of a passenger vehicle (approximately **6 ft** wide):

$$173 \text{ pixels} \div 6 \text{ ft} = 28.8 \text{ pixels/ft}$$

28.8 pixels/ft exceeds the recommended **20–25 pixels/ft** design target for ITS. This supports signal actuation, queue detection, vehicle classification, wrong-way detection, and speed estimation.

Rule of Thumb: Measure a known object (6-ft vehicle or 12-ft lane) and calculate **Pixels ÷ Actual Width = Pixels/ft**.